

RESEARCH VISION

My research vision is to build **provably safe, verifiable, and trustworthy AI systems**. I bridge formal language theory and modern AI/NLP, applying theoretical tools—formal grammars and automata—to create AI systems with **mathematical guarantees** on safety and correctness.

- **Core Approach: Bridging Formal Language Theory and Modern AI**

My Ph.D. research bridges formal language theory and AI/NLP to address this challenge. I advance foundational questions in computational theory—such as decidability of string properties (Parikh matrices), language hierarchies, and formal characterization of structural constraints—and directly apply these theoretical tools to build **verifiable AI systems**. By integrating symbolic, grammar-based methods with neural models, I create frameworks where AI behavior can be formally specified and mathematically verified.

- **Future Focus: Verifiable LLM Safety**

I aim to apply these methodologies to the most pressing AI challenge: **provably safe LLMs**. By formalizing safety constraints as decidable properties in formal language theory, I will develop systems with **mathematical verification** of outputs, guaranteed abstention mechanisms, and transparent specifications that enable external auditing. My unique expertise in both formal methods and modern NLP positions me to pioneer **verifiable AI safety**—moving beyond heuristics to mathematical guarantees.

EDUCATION

Yonsei University

Integrated Ph.D. course in Computer Science
Supervisor: Yo-Sub Han

Seoul, KR
2019–Current

Thesis: *Symbolic Techniques for Deep Learning in Natural Language Processing and Program Analysis*

Yonsei University

Bachelor's degree in Computer Science

Seoul, KR
2015–2019

PUBLICATIONS

Full list: [\[Google Scholar\]](#) [\[DBLP\]](#)

† indicates equal contribution

Peer-Reviewed Conference/Journal Papers

- [C1] **EnCur: Curriculum-Based In-Context Learning with Structural Encoding for Code Time Complexity Prediction**

Joonghyuk Hahn, Aditi, Seung-Yeop Baik, Shinwoo Park, Sang-Ki Ko, Yo-Sub Han
To appear in ESWA 2026 [\[PDF\]](#)

- [C2] **AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection**

Yejin Lee, Joonghyuk Hahn, Hyeseon Ahn, Yo-Sub Han
In Proceedings of EMNLP 2025 [\[PDF\]](#)

- [C3] **CodeComplex: Dataset for Worst-Case Time Complexity Prediction**

Seung-Yeop Baik, Joonghyuk Hahn, Jungin Kim, Mingi Jeon, Aditi, Yo-Sub Han, Sang-Ki Ko
In Findings of EMNLP 2025 [\[PDF\]](#)

- [C4] **TCProF: Time-Complexity Prediction SSL Framework**

Joonghyuk Hahn, Hyeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han
In Proceedings of NAACL 2025 [\[PDF\]](#)

- [C5] **Advanced Code Time Complexity Prediction Approach Using Contrastive Learning**
Shinwoo Park, **Joonghyuk Hahn**, Elizabeth Orwig, Sang-Ki Ko, Yo-Sub Han
In EAAI 2025, Vol. 151 [\[PDF\]](#)
- [C6] **Characterizations of M-equivalence and weak M-relation**
Joonghyuk Hahn, Hyunjoon Cheon, Yo-Sub Han
In IJFCS 2025 [\[PDF\]](#)
- [C7] **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], **Joonghyuk Hahn**[†], Yo-Sub Han
In Theory of Computing Systems 2024, Vol. 68, 301–321 [\[PDF\]](#)
- [C8] **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem**
Ingyu Baek[†], **Joonghyuk Hahn**[†], Yo-Sub Han, Kai Salomaa
In Proceedings of DLT 2024, LNCS 14791, 68–81 [\[PDF\]](#)
- [C9] **SuperST: Superficial Self-Training for Few-Shot Text Classification**
Ju-Hyoung Lee[†], **Joonghyuk Hahn**[†], Hyeon-Tae Seo, Jiho Park, Yo-Sub Han
In Proceedings of LREC-COLING 2024 [\[PDF\]](#)
- [C10] **ATHENA: Mathematical Reasoning with Thought Expansion**
J.B. Kim, Hazel Kim, **Joonghyuk Hahn**, Yo-Sub Han
In Proceedings of EMNLP 2023, 16315–16327 [\[PDF\]](#)
- [C11] **GDA: Grammar-based Data Augmentation for Text Classification using Slot Information**
Joonghyuk Hahn, Hyunjoon Cheon, Elizabeth G. Orwig, Su-Hyeon Kim, Sang-Ki Ko, Yo-Sub Han
In Findings of EMNLP 2023 [\[PDF\]](#)
- [C12] **M-equivalence of Parikh Matrix over a Ternary Alphabet**
Joonghyuk Hahn, Hyunjoon Cheon, Yo-Sub Han
In Proceedings of CIAA 2023 [\[PDF\]](#)
- [C13] **Boosting Code Summarization by Embedding Code Structures**
Jikyeong Son[†], **Joonghyuk Hahn**[†], Hyeon-Tae Seo, Yo-Sub Han
In Proceedings of COLING 2022, 5966–5977 [\[PDF\]](#)
- [C14] **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], **Joonghyuk Hahn**[†], Yo-Sub Han
In Proceedings of DLT 2022, LNCS 13257, 115–126 [\[PDF\]](#)
- [C15] **Self-Training using Rules of Grammar for Few-Shot NLU**
Joonghyuk Hahn, Hyunjoon Cheon, Kyuyeol Han, Cheongjae Lee, Junseok Kim, Yo-Sub Han
In Findings of EMNLP 2021 [\[PDF\]](#)
- [C16] **Most Pseudo-copy Languages Are Not Context-Free**
Hyunjoon Cheon[†], **Joonghyuk Hahn**[†], Yo-Sub Han, Sang-Ki Ko
In Proceedings of COCOON 2021, 189–200 [\[PDF\]](#)

Under Review

- **MEC³O: Multi-Expert Consensus for Code Time Complexity Prediction**
Joonghyuk Hahn[†], Soohan Lim[†], Yo-Sub Han
Under review
- **Query4Regex: Verifiable Regex Transformation through Formal Operations from NL and DSL Queries**
Joonghyuk Hahn, Yo-Sub Han
Under review
- **Repairing Regex Vulnerabilities via Localization-Guided Instructions**
Sicheol Sung[†], **Joonghyuk Hahn**[†], Yo-Sub Han
Under review
- **A Regex Minimization Benchmark: A PSPACE-Complete Challenge for Language Models**
Hyundong Jin, **Joonghyuk Hahn**, Yo-Sub Han
Under review

- **ECO: Enhanced Code Optimization via Performance-Aware Prompting for Code-LLMs**
Su-Hyeon Kim, Joonghyuk Hahn, Sooyung Cha, Yo-Sub Han
Under review
- **A Voronoi-Embedding Framework for Semi-Supervised Time-Complexity Prediction**
Hyeseon Ahn, Joonghyuk Hahn, Yo-Sub Han
Under review

PROJECTS

2023–2025: Human-AI Programming Platform Research (follow-up)

Ministry of Science and ICT

- Efficient Frameworks for programming platform with human and AI collaboration.

2022–2022: 5th AI Grand Challenge

Ministry of Science and ICT

- Developed a deep learning model for solving mathematical equation problems in a three-stage national-level competition.
- Passed the first stage and wrote the proposal for the competition projects. Then, participated in the second-stage competition.

2019–2022: Malware Pattern Extraction & Human-AI Programming Platform Research

Ministry of Science and ICT

- Researched synergies between NLP and source code, such as code summarization and complexity analysis.

RESEARCH INTERESTS FOR FUTURE WORKS

- **Mechanistic Interpretability of Generative Models:** Applying formal language theory and automata to understand internal mechanisms of LLMs—analyzing learned circuits, feature representations, and reasoning processes to make models more transparent and interpretable.
- **Formal Reasoning & Symbolic Integration:** Bridging symbolic methods (formal grammars, logical constraints, decidability analysis) with neural models to enable verifiable reasoning, compositional generalization, and systematic problem-solving in language models.
- **Constrained & Controllable Generation:** Developing grammar-based frameworks for structured, safe generation with guarantees on output properties—applicable to code synthesis, mathematical reasoning, and domain-specific language generation.
- **Efficient & Robust Language Models:** Investigating byte-level models, robustness to perturbations, and efficient architectures informed by formal language hierarchies—advancing foundations for more resilient and efficient generative systems.
- **Neuro-Symbolic AI Safety:** Bridging symbolic reasoning (formal methods, logical constraints) with neural models to create interpretable, verifiable AI systems with transparent failure modes.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java
- **ML/NLP Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers
- **Formal Methods:** Automata theory, formal grammars (CFG, PDA), decidability analysis, formal language hierarchies
- **Research Areas:** Mechanistic interpretability, constrained generation, neuro-symbolic AI, formal verification for neural networks
- **Tools:** L^AT_EX, Git, Linux/Unix

REFERENCES

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